MPGV Monitor – Stress Test Plan (Pluto-CSHIS-9001)
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Goal: Test how many concurrent users the deployed MPGV Monitor can handle on pluto.cs.hi.is:9001 – including real-time map usage, timeline slider, toggles, and API calls.

1. Objectives

• Evaluate behavior under 20–40+ users.
• Measure response times, throughput, errors, CPU/RAM.
• Identify bottlenecks (Gunicorn, SQLite, frontend rendering).

2. Test Environment

Server: http://pluto.cs.hi.is:9001/
Frontend load: map zoom/pan, slider zoom, toggles.
Backend load: /earthquakes, /update-earthquakes, /reconcile.

3. Volunteer Instructions

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1. Open the site.
2. Interact for 3–5 minutes:
– Zoom map
– Drag map
– Move timeline slider
– Toggle layers
– Change map type
– Refresh sometimes
3. Fill feedback form afterward.

4. Backend Load Testing (Instructor)

ApacheBench examples:
ab -n 200 -c 20 http://pluto.cs.hi.is:9001/earthquakes
ab -n 1000 -c 100 http://pluto.cs.hi.is:9001/earthquakes

5. Monitoring Pluto

Use: top, vmstat, tail -f logs.

6. Recording Results

Save ab outputs, system snapshots, form feedback.

7. Interpretation

Good: low latency, no errors.
Bad: slow responses, CPU 100%, DB lock issues.